



Machine learning-based estimation of gaseous and particulate emissions using internally observable vehicle operating parameters

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ABSTRACT

Measuring vehicular emissions is crucial for emission management and air quality control. However, conventional measurement equipment is costly and requires continuous maintenance. This study aims to address these challenges by introducing an Artificial Neural Network (ANN)-based methodology that using internally observable vehicle operating parameters to estimate emissions of particulate number (PN), nitrogen oxide (NO_x), carbon monoxide (CO), and carbon dioxide (CO₂). The research was conducted in three steps: (1) analyzing the correlations between vehicle operating parameters and exhaust emissions through emission map analysis, (2) developing and validating artificial neural networks (ANNs), and (3) examining the impact of vehicle driving conditions on emission predictions. The ANNs were rigorously validated across various operating and ambient conditions, including different temperatures, cold and warmed-up starts, and various driving scenarios. The prediction accuracy for CO₂ is higher than other exhaust gases, with an error of <1% in all validation cases. The mean prediction error for NO_x was 12.5%, with observed variations in accuracy depending on dynamic and thermal conditions. The ANN model accurately predicted the influence of thermal conditions on CO emissions, which exhibit a significant increase at lower temperatures.

1. Introduction

Vehicle emissions are a significant sources of greenhouse gases and atmospheric pollutants (Colville et al., 2001; Fan et al., 2018). In response to stringent regulations aimed at reducing environmental impact (Johnson and Joshi, 2018), modern vehicles are now equipped with advanced engine and emission control technologies. These include high-pressure fuel injection, advanced combustion, variable valve timing control, and cooled exhaust gas recirculation, coupled with aftertreatment devices such as particulate filters and catalytic converters (Conway et al., 2021; Joshi, 2022).

When advanced engine and emission control technologies are applied to vehicles, exhaust emissions are affected by numerous factors, resulting in complicated emission characteristics. Emissions are initially generated within the engine and are modified by

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aftertreatment devices before being released into the atmosphere. These modifications depend on both engine operating conditions such as thermal state, mixture formation, and engine load (Ahire et al., 2021; Dardiotis et al., 2013), as well as the performance of the aftertreatment devices (Jiang and Li, 2016; Gao et al., 2019). Therefore, pollutant emission levels are influenced by various operating parameters, such as vehicle speed, engine speed, engine load, engine temperature, and the conversion efficiency of aftertreatment devices.

In addition to the operating conditions of the vehicle, exhaust emissions are closely related to environmental factors. In on-road driving, uncontrollable factors such as various speed patterns resulting from the interaction with surrounding vehicles and traffic signals, ambient temperature, and road grade affect exhaust emissions. The recent introduction of real driving emission (RDE) regulations has led to increased demand for research that takes these environmental factors into account (Bielaczyc and Woodburn, 2019; Fontaras et al., 2017).

As mentioned above, the characteristics of exhaust emissions are becoming more complicated due to advances in emission control technologies and the influence of on-road driving conditions. In terms of air quality controls, understanding these complexities is essential for accurate emission assessment and prediction. However, conventional emission models often fall short in predicting instantaneous gaseous and particulate emissions.

Emission inventory models based on simple correlations of experimental data have been widely used to estimate vehicle exhaust emissions (Ntziachristos et al., 2009; Vallamsundar and Lin, 2011; Koupal et al., 2002; Smit et al., 2007). Those emission models express exhaust emissions as a function of simple vehicle operating parameters, such as the average or instantaneous vehicle speed, vehicle specific power, traffic density, and road gradient. In addition, modal emission models, which are based on empirical correlations between vehicle operating parameters and exhaust emissions, are sometimes used (Smit et al., 2010; Hausberger, 2009). However, these methods often are not sufficient for accurately predicting instantaneous emissions.

In contrast, analytical emission models incorporate detailed emission generation mechanisms and computational fluid dynamics to estimate exhaust emissions with high prediction accuracy (Kumar et al., 2013; Zhao et al., 2017). These models consider physical phenomena such as fuel spray, ignition delay, diffused and premixed combustion, heat transfer, and pollutant formation. However, they require significant computational resources, advanced modeling techniques, and detailed design parameters impractical for assessing emissions from a broad range of vehicles.

Much research in recent years has focused on using artificial intelligence, specifically artificial neural networks (ANNs), for vehicle emission modeling. ANNs are well-suited for this task because they can handle the complex physical processes involved (Meireles et al., 2003; Sharma et al., 2005). Capable of recognizing intricate emission patterns through learning, ANNs eliminate the need for detailed knowledge or numerical calculations in estimating emissions. As a result, ANN-based approaches are cost-effective and adaptable, offering a practical alternative to conventional emission models (Khurana et al., 2021).

ANN-based emission models have been extensively employed to estimate vehicle pollutants. Jaikumar et al. (2017) developed an ANN-based carbon monoxide (CO), hydrocarbon (HC), and nitrogen oxide (NOx) prediction model that used vehicle speed, acceleration, engine speed, and vehicle specific power as the model inputs. Their validation results show that the prediction accuracy of the ANN was high under various driving conditions. Moradi and Miranda-Moreno (2020) used machine learning to estimate fuel consumption and achieved satisfactory accuracy even when using low-cost devices for data collection. Goulier et al. (2020) focused on estimating gaseous and particulate concentrations in urban areas and incorporated meteorological variables such as temperature and humidity as inputs. While their model was successful in predicting gaseous emissions, it was less accurate for particulate emissions. Alfaseeh et al. (2020) utilized various neural network architectures, specifically long-short term memory (LSTM), to forecast greenhouse gas emissions in urban areas, and found that increasing the network depth enhanced prediction performance. Fei et al. (2023) introduced a hybrid model using temporal convolutional networks (TCN), convolutional neural networks (CNN), and LSTM to deal with the challenges of non-stationarity and nonlinearity in emission prediction. Nam et al. (2021) provided a national-scale prediction of polycyclic aromatic hydrocarbons, employing deep learning techniques and multimedia fugacity models for greater accuracy. Loy-Benitez et al. (2022) developed a data-based soft-sensor to predict indoor PM_{2.5} levels using outdoor and indoor air variables. The approach incorporates neural circuit policies and nature-inspired liquid time-constant networks. This method demonstrates improved predictive accuracy compared to other neural methods. Rivera-Campoverde et al. (2021) estimated vehicular carbon dioxide (CO₂), NOx, CO, and HC emissions using an ANN trained with on-road test data. Various parameters, such as throttle position, mass air flow, manifold pressure, intake air temperature, engine coolant temperature, vehicle speed, engine speed, acceleration, fuel flow, and GPS information, were considered as predictors in their model. Although the proposed model showed high accuracy in predicting gaseous emissions, it did not perform well in predicting particulate emissions.

Thus, although many studies have estimated gaseous emissions from vehicles, few researchers have proposed particulate emission models. Therefore, we introduce here novel and feasible ways to estimate both gaseous and particulate vehicle emissions based on ANNs. The proposed emission models were designed to predict instantaneous NOx, CO, CO₂, and particulate number (PN), and we rigorously validated the emission prediction accuracy under various conditions: low/high ambient temperature, cold/warmed-up start, and urban/rural/motorway driving. In addition, we analyzed the effects of driving conditions, such as vehicle speed and engine temperature, on emission predictions.

2. Methodology

2.1. Vehicle test data

During the on-road test, a comprehensive range of variables was collected, including vehicle speed, engine speed, engine load,

engine coolant temperature, intake air mass flow, and NO_x, CO, CO₂, and PN emissions. Internally observable variables (IOVs) were measured once per second during the test using an onboard diagnostics channel, and instantaneous exhaust emissions were measured using a portable emission measurement system (PEMS) (Semtech Ecstar, Sensors, Michigan, USA). The exhaust gas analyzer on the PEMS measured CO₂ (up to 18 vol%) and CO (up to 8 vol%) using heated non-dispersive infrared detection. The NO_x measurement used a non-dispersive ultraviolet analyzer (up to 3000 ppm). For PN measurement, a micro-proportional sampling system was used to dilute the exhaust gas, and then a continuous particulate counter measured the PN emissions with a size from 23 nm to 2.5 µm. The exhaust gas flow meter could measure a mass flow rate of up to 670 kg/h. To ensure the reliability of the PEMS, we compared the PEMS-measured data with test results from a certified exhaust gas analyzer from the National Institute of Environmental Research, which is a national agency for emission certification. The specifications of PEMS used in this study are described in previous studies (Park et al., 2021; Cha et al., 2019).

The emission characteristics of vehicle exhaust are sensitive to vehicle operating and environmental conditions and differ by gas type. The correlations between the measured variables and exhaust emissions are complex, and certain variables are more closely related to exhaust emissions than others. For example, CO₂ emissions tend to increase as the vehicle speed increases, and typically, an engine that is not fully warmed-up (as shown by the engine coolant temperature) emits more HC and CO than a warmed-up engine. Using machine learning techniques, ANNs can be trained on the emission characteristics from on-road test data. In this study, we trained ANNs with on-road test data to predict exhaust emissions. To develop an emission model that can accurately predict exhaust emissions under various conditions, it is important to train the ANN with a sufficient amount of training data containing various emission characteristics. Both the quality and quantity of the training data directly affect the performance of an ANN. In terms of the quality of the training data, on-road test data measured under various conditions, such as urban/rural/motorway driving, low/high ambient temperature, and cold start/warmed-up engine start, are more favorable for ANN training than test data measured under static conditions. Because an ANN cannot consider conditions not contained within the training data, insufficient training data can degrade ANN performance. In addition, overfitting is more likely to occur when the amount of training data is insufficient.

We used a massive amount of test data from a diesel passenger vehicle to develop our ANN-based emission prediction model. The vehicle test conditions are shown in Supplementary Table S1. 23 on-road tests (total 39 h) were performed around Seoul and in Incheon Province, and details on the test routes can be found in Supplementary Fig. S1 and Table S2. The test routes and test result of this study comply with the requirements for real-world driving test in South Korea (Park et al., 2021). According to the real-world driving test regulations, the driving route should consist of urban, rural, and motorway driving in the proportions of 34%, 33%, and 33% respectively (with an error margin of ±10%). Urban driving refers to driving at speeds below 60 km/h, rural driving refers to speeds ranging from above 60 km/h to 90 km/h, and motorway driving refers to conditions where the vehicle's speed exceeds 90 km/h. However, the speed ranges may be adjusted based on the performance of the vehicle. The stationary period is defined as a vehicle speed below 1 km/h, and it should occupy 6–30% of the urban driving time. Urban driving may include stationary periods lasting >10 s, but if the individual stationary periods exceed 300 s continuously, the test is considered invalid. Additionally, the altitude difference between the starting and ending points should not exceed 100 m, and the driving distances for urban, rural, and motorway sections should each be at least 16 km.

Thirteen tests were conducted under cold start conditions. In the remaining ten tests, exhaust emissions were measured only after the engine was fully warmed-up. As shown in Supplementary Table S3, the vehicle tests were conducted from August to December, and the test were carried out between 09:00 and 19:00. The lowest and highest mean ambient temperatures during the tests were −2.3 °C and 34.1 °C, respectively. In addition, vehicle tests were performed on urban, rural, and motorway routes near Seoul to accumulate data under various operating conditions of acceleration and deceleration, cruising, and traffic congestion. The highest measured vehicle speed was 133 km/h. Supplementary Fig. S2 shows the distribution of measured vehicle speed (1 Hz) data. The bin size was 5 km/h, and 141,500 data points were collected. The frequency of the 0–5 km/h bin is the highest because it includes idling operation. The high frequency of the 45–60 and 75–85 km/h speed ranges reflect the speed limits. Typical speed limits for road vehicles in urban and rural areas of Korea are 50–60 km/h and 80–90 km/h, respectively.

Supplementary Table S4 shows the major specifications of the test vehicle, a 2017 model-year diesel passenger vehicle subject to the Euro 6d-temp emission standard, which includes both the worldwide harmonized light-duty vehicles test procedure and RDE regulations. The emission standards of the Euro 6d-temp regulation are more strict than those of previous regulations, such as Euro 5 and Euro 6b. Typically, most vehicles covered by Euro 6d-temp regulation are equipped with aftertreatment systems to meet the strict emission limits (Valverde Morales et al., 2020; Joshi, 2022; Bielaczyc and Woodburn, 2019). The test vehicle in this study was equipped with a diesel particulate filter (DPF), a lean NO_x trap (LNT), and urea-selective catalytic reduction (urea-SCR). Engine-generated emissions were emitted into the atmosphere after passing through the LNT, DPF, and urea-SCR in that order. The main function of the DPF is the reduction of particulates from exhaust gases, and the LNT and urea-SCR are designed to reduce NO_x

Table 1
Experimental conditions of the five test runs selected for ANN validation.

	Starting condition	Average ambient temperature
Test 1	Warmed up	6.3 °C
Test 2	Warmed up	28.9 °C
Test 3	Cold	4.7 °C
Test 4	Cold	26.6 °C
Test 5	Cold	29.2 °C

emissions (Forzatti et al., 2010).

We separated the experimental data into training and validation sets so that the data already used for ANN training would not be reused in the prediction accuracy evaluation. We selected five test runs for the accuracy evaluation from among the 23 on-road test runs completed. The experimental conditions for the five test runs are shown in Table 1 and Fig. 1. To rigorously validate the performance of the ANNs, we evaluated their emission prediction accuracy under different driving conditions such as warmed-up starting, cold starting, high temperature, low temperature, urban driving, rural driving, and highway driving. Except for the five test runs for accuracy validation, the rest of the data were used to train the ANNs by adjusting weighting factors or monitoring for overfitting. Among these data, 10% was randomly allocated for the purpose of monitoring overfitting, while the remaining 90% was utilized for training the ANNs.

2.2. Development of the artificial neural network

Before training the ANNs with on-road test data, we analyzed the relationship between the measured vehicle operating parameters and exhaust emissions using emissions maps. These maps plotted vehicle operating parameters on the x and y axes, and emission rates on the z axis.

We developed three distinct types of emission maps: Type 1 considered vehicle speed, intake air mass flow, and emission rates; Type 2 focused on engine speed, fuel-air equivalence ratio, and emission rates; Type 3 looked at engine coolant temperature, intake air temperature, and emission rates. These emission maps each used two vehicle operating variables to separately represent CO₂, NO_x, CO, and PN emissions. The Type 1 and Type 2 maps were designed to analyze the effects of dynamic conditions on exhaust emissions. In contrast, the Type 3 map was developed to show the correlation between thermal conditions and exhaust emissions.

We used on-road test data and statistical data binning to develop these maps. The on-road emission data were allocated into a two-dimensional grid of vehicle operating parameters. The intervals of vehicle speed, intake air mass flow, engine speed, fuel-air equivalence ratio, engine coolant temperature and intake air temperature were 10 km/h, 10 g/s, 200 rpm, 0.2, 10 °C, and 10 °C, respectively. The representative emission rate of each group was the arithmetic mean of exhaust emissions belonging to that group.

We developed a multi-layer neural network to predict vehicle exhaust emissions based on our previous study. (Seo and Park, 2023). The model was designed to predict the emissions generated per second by using the vehicle operating parameters for the preceding and subsequent 2 s of the desired prediction second. This short history of input data allows the model to account for transient effects, such

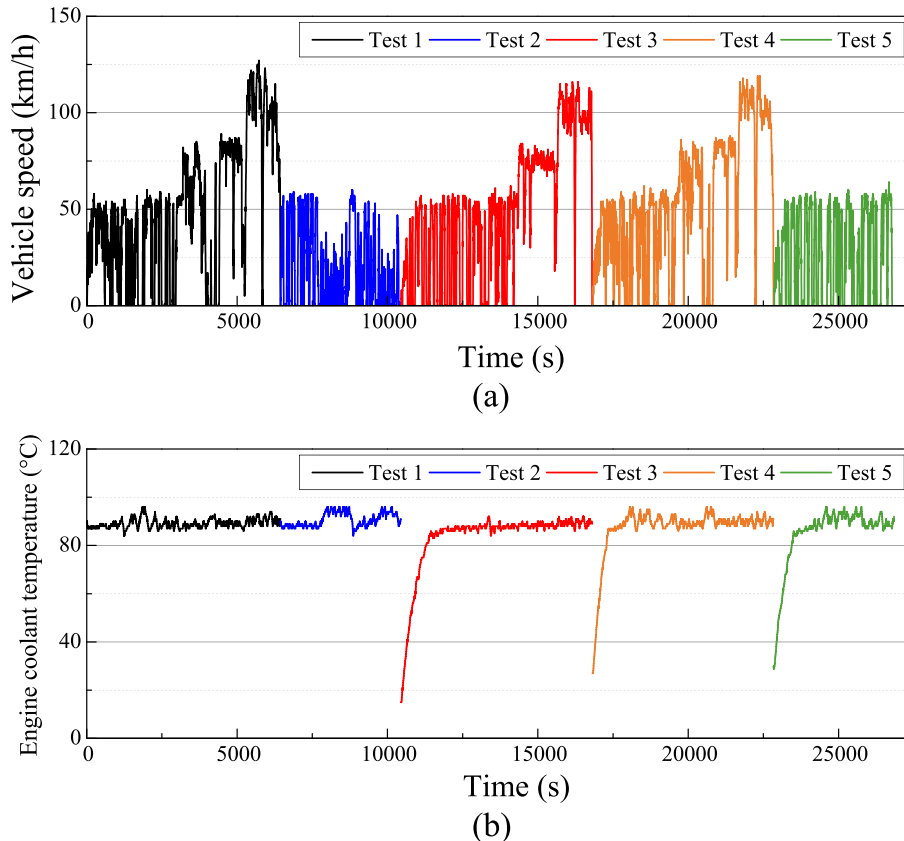


Fig. 1. Experimental conditions of the five test runs used for ANN validation: (a) vehicle speed; (b) engine coolant temperature.

as signal distortions and measurement delays, which are particularly significant in on-road tests. (Franco García, 2014; Ma et al., 2020). Zhang and Frey (2008) evaluated that time mismatches in PEMS systems were reported to be approximately 3 s for CO₂, 6 s for NO, 3 s for CO, and 3 s for HC emissions. To address this, our ANN incorporates 5 s of vehicle driving data around the desired prediction second, enabling the ANN to directly compensate for transient effects in the experimental data. Since vehicle emissions measurements usually collect continuous data streams, utilizing information for periods subsequent to the desired prediction second presents no significant challenge. In our previous work, (Seo and Park, 2023), this approach reduced the prediction root mean square error (RMSE) by up to 45% compared to models using only 1 s of vehicle operating parameters.

Fig. 2 shows the ANN structure employed in this study. Each ANN is designed to predict instantaneous exhaust emissions using selected vehicle operating parameters as input variables. Out of the many parameters measured during the driving test, we choose seven parameters that are highly correlated with exhaust gas emissions through trial and error. The output variables of the ANNs were instantaneous NO_x, CO, CO₂, and PN emissions. Separate ANNs were developed to predict each type of exhaust emission.

Among the input variables for the ANNs, vehicle speed is a basic measure of the vehicle's operating condition and is easily obtainable. Engine speed directly indicates the operating state of the engine. The engine coolant temperature reflects the engine's thermal condition, which significantly affects exhaust emissions. Before the engine is fully warmed-up, it tends to emit more pollutants due to deteriorated evaporation of fuel and incomplete combustion (Weilenmann et al., 2009). The fuel-air equivalence ratio represents the oxidizer proportion of oxidizer, affecting both engine efficiency and exhaust gas chemistry. Most modern vehicles finely control this ratio to optimize energy efficiency and minimize emissions (Brijesh and Sreedhara, 2013). The exhaust gas temperature reflects the performance of the aftertreatment devices, which use temperature-sensitive catalysts whose performance deteriorates below a certain temperature (Jiang and Li, 2016). Intake air mass flow and intake air temperature are closely related to engine performance and exhaust gas characteristics (Cinar et al., 2015). The density of air is indirectly proportional to its temperature, and high air density increases the volumetric efficiency of the engine.

Each of our ANNs consists of an input layer, an output layer, and five hidden layers, as shown in Supplementary Table S5. The RMSprop algorithm was used as optimizes, adjusting learning rate for each weight based on the magnitude of the gradients through an exponentially weighted moving average. Relu and the RMSE were used as the activation function and loss function, respectively. During training, no signal of overfitting were observed. The ANN was implemented using Python 3.9 and utilized Tensorflow 2.0 for machine learning tasks.

To effectively train the ANNs, the input variables were standardized to ensure a common scale, mitigating the risk of overestimation or underestimation due to varying scales. To maintain independence between the training and validation datasets, standardization was performed separately for each. The standardization (STD) can be described as follows:

$$STD(x) = \frac{x - MEAN(x)}{SD(x)} \quad (1)$$

where x is the standardized variable, $MEAN(x)$ is the mean of x , and $SD(x)$ is the standard deviation of x .

Log-transformation was applied to PN data. Because the wide distribution of PN values, ranging from 0 to 10¹² particles/s, training the ANN on the original PN data became dominated by large values, leading to low and unstable prediction accuracy. Log-transformation, specifically using log base 10, substantially improved training efficiency by creating a more uniform distribution of the data. In this process, only PN emissions higher than 10 were log-transformed, while those lower than 10 were set to zero.

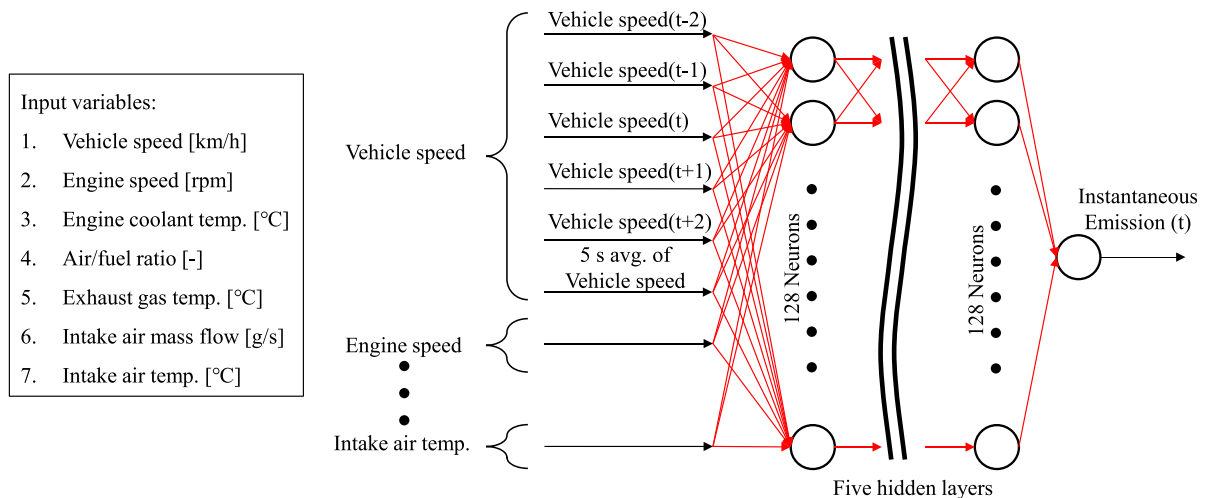


Fig. 2. Input variables and structure of the artificial neural networks.

3. Results

3.1. Emission maps

Before developing our ANN-based emission models, we analyzed the relationships between the vehicle operating parameters and exhaust emissions using emission maps. We used the emission maps to visually describe how gaseous and particulate emissions varied with vehicle operating parameters (engine speed, fuel-air equivalence ratio, engine coolant temperature, and intake air temperature). Those operating parameters were also used as model inputs for the ANNs.

Three types of emission maps were developed using on-road emission data and statistical data binning. Fig. 3 shows the exhaust emissions according to vehicle speed and intake air mass flow. The NO_x, CO, and CO₂ emissions all increased as the vehicle speed and intake air mass flow increased, as shown in Fig. 3(a), (b), and (c). Their emission rates were thus highest at high speed and a high intake air mass flow area. In contrast, Fig. 3(d) illustrates that PN emissions are not proportional to vehicle speed and intake air mass flow, and no consistent pattern is visible.

Fig. 4 shows the exhaust emissions according to the engine speed and fuel-air equivalence ratio. As shown in Fig. 4(a), the fuel-air equivalence ratio had lesser effect on NO_x emissions than engine speed, and NO_x emissions increased as the engine speed increased. In contrast, CO emissions were significantly sensitive to the fuel-air equivalence ratio, as shown in Fig. 4(b). When the fuel-air equivalence ratio was close to 1, CO emissions were quite high, whereas in other regions, the CO emissions were low. The effect of engine speed on CO emissions was quite small. In the CO₂ emission map in Fig. 4(c), it is clear that CO₂ emissions increased gradually as the engine speed and fuel-air equivalence ratio increased. Because CO₂ emissions were proportional to fuel consumption, they were highly correlated with the fuel-air equivalence ratio. The PN emission characteristics were similar to those of CO. As shown in Fig. 4(d), PN emissions were high when the fuel-air equivalence ratio was close to 1. It is important to note that a logarithmic scale is used to express the PN emission rates.

Fig. 5 illustrates the effect of thermal conditions on exhaust emissions. Due to limited data at engine coolant temperatures below 20 °C, the emission map only displays engine coolant temperatures above 20 °C. NO_x and CO emissions were sensitive to the engine

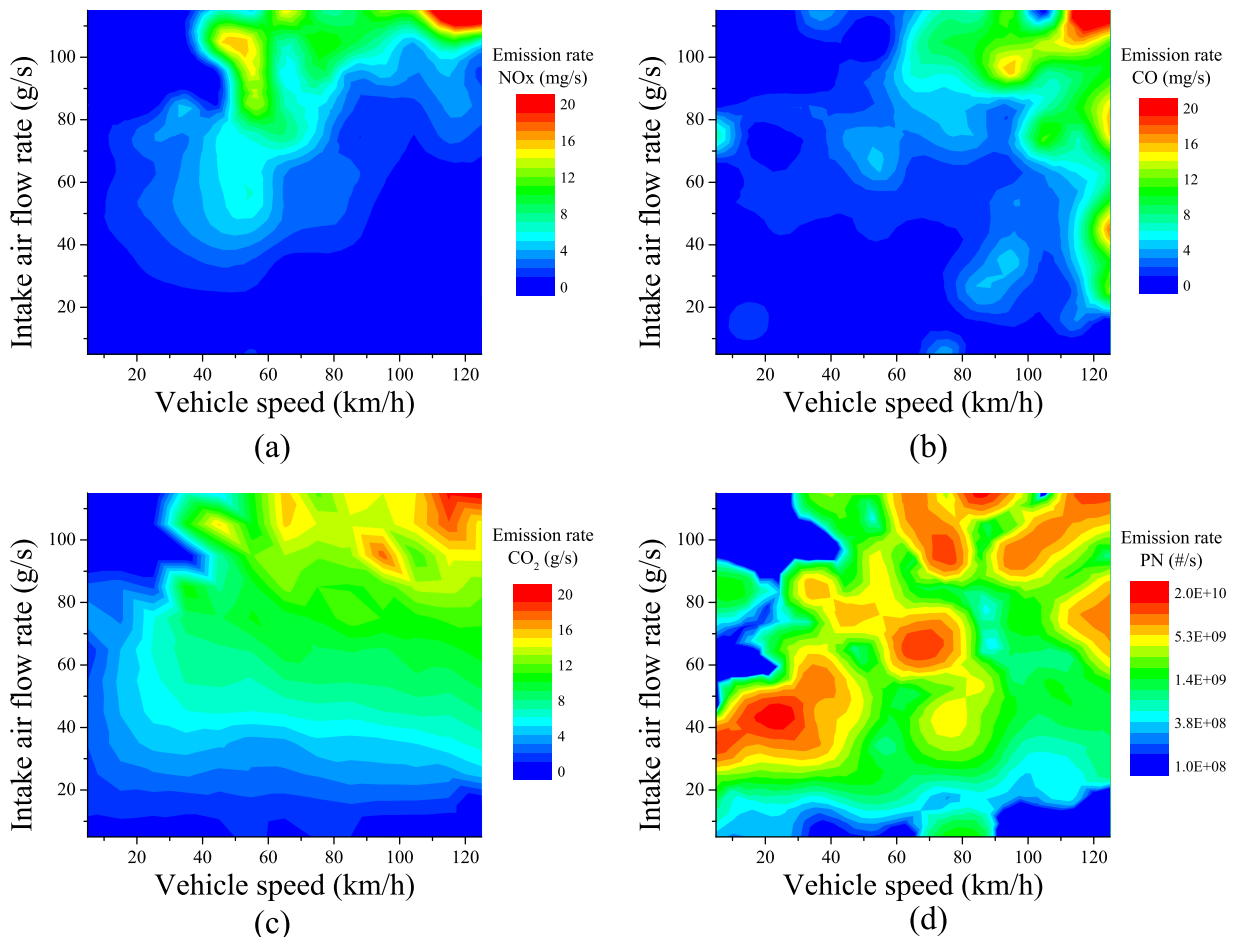


Fig. 3. Exhaust emissions according to vehicle speed and intake air mass flow: (a) NO_x; (b) CO; (c) CO₂; (d) PN.

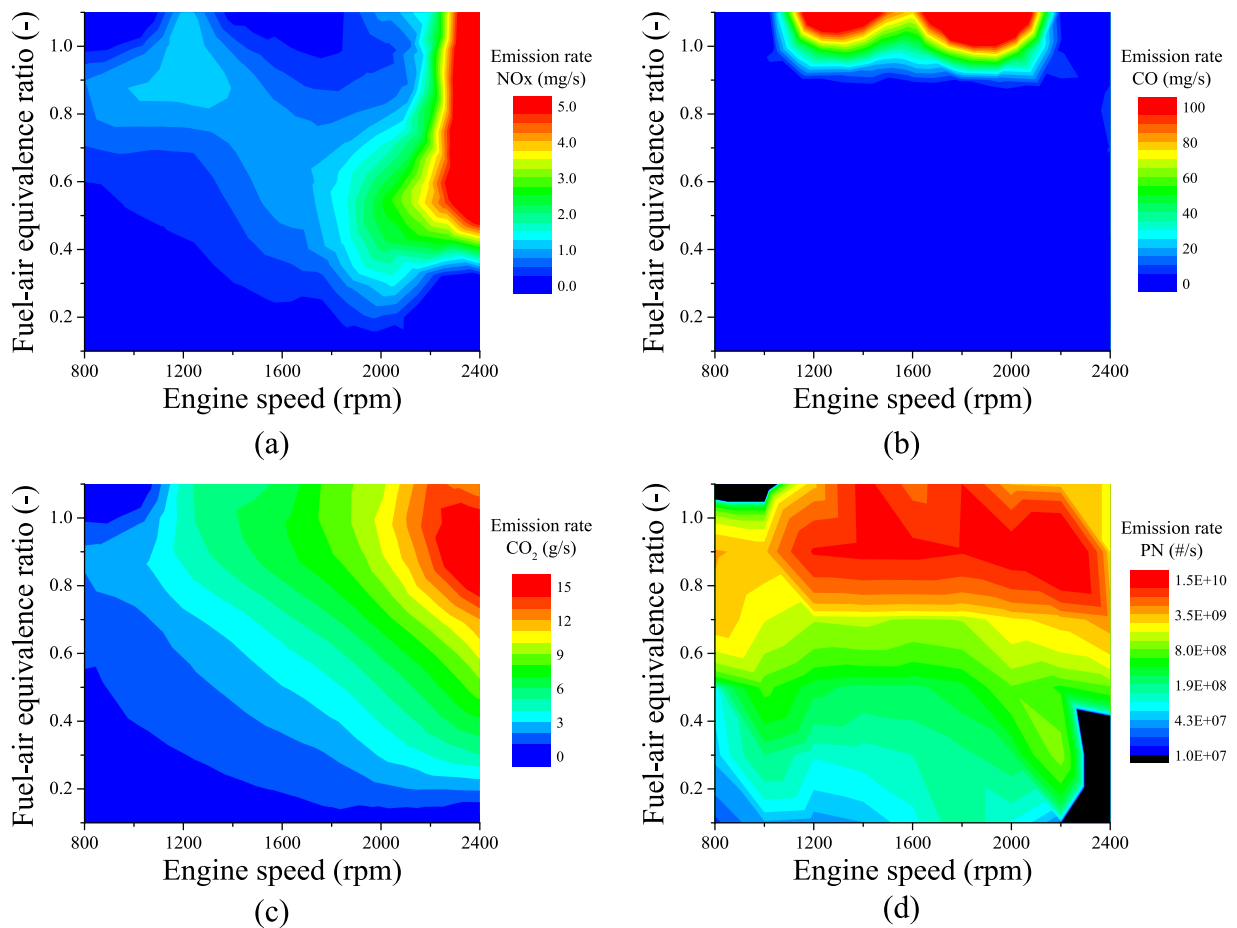


Fig. 4. Exhaust emissions according to the engine speed and fuel-air equivalence ratio: (a) NO_x; (b) CO; (c) CO₂; (d) PN.

coolant temperature. As shown in Fig. 5(a) and (b), the NO_x and CO emissions were high when the engine coolant temperature was low. The CO emissions were also sensitive to the intake air temperature and were highest when both the engine coolant temperature and intake air temperature were low. In contrast, as shown in Fig. 5(c), CO₂ emissions were not high when the engine coolant and intake air temperatures were low. Most areas of this graph show CO₂ emissions of around 2 g/s (green), but they increased to about 3 g/s (red) when the engine coolant temperature was above 90 °C. Fig. 5(d) shows that PN emissions were high when both the engine coolant temperature and intake air temperature were high. Although it is not clear in Fig. 5(d), the PN emissions were also high when the engine coolant temperature was low, as shown in Supplementary Fig. S3.

3.2. ANN-based emission prediction results

3.2.1. NO_x emissions

In this study, we validated the prediction accuracy of the ANNs by comparing their predicted emissions with the measured emissions. As mentioned in the methodology section, the five test runs in Table 1 were not used to train the ANNs but saved to use in the evaluation of the emission prediction accuracy. The comparison results for the instantaneous NO_x emission rates and cumulative NO_x emissions are illustrated in Supplementary Figs. S4 and S5, respectively.

High NO_x emission rates (in excess of 30 mg/s) are shown only in cold weather conditions in Fig. S4(a) and (c). In contrast, instantaneous NO_x emissions exceeding 20 mg/s rarely occurred in warm conditions, as shown in Supplementary Fig. S4(b), (d), and (e). The thermal condition of the vehicle affected NO_x emissions immediately after starting. Under cold start conditions, a slight increase in NO_x emissions is shown in Supplementary Fig. S4(d) and (e). In the cumulative NO_x emissions shown in Supplementary Fig. S5(d) and (e), more NO_x was emitted in the first 300 s after starting than in the other sections. This tendency is also visible in the NO_x emission map in Fig. 5(a), where the NO_x emissions are high when the engine coolant temperature is low.

Supplementary Fig. S4 shows that, the ANN accurately predicts the points at which NO_x emissions increase or decrease. However, as shown in Supplementary Fig. S5, the cumulative emissions, which are indicative of the overall NO_x emission levels predicted by the ANN, exhibit some degree of error. Therefore, when utilizing the ANN for NO_x prediction, it is anticipated to be useful for

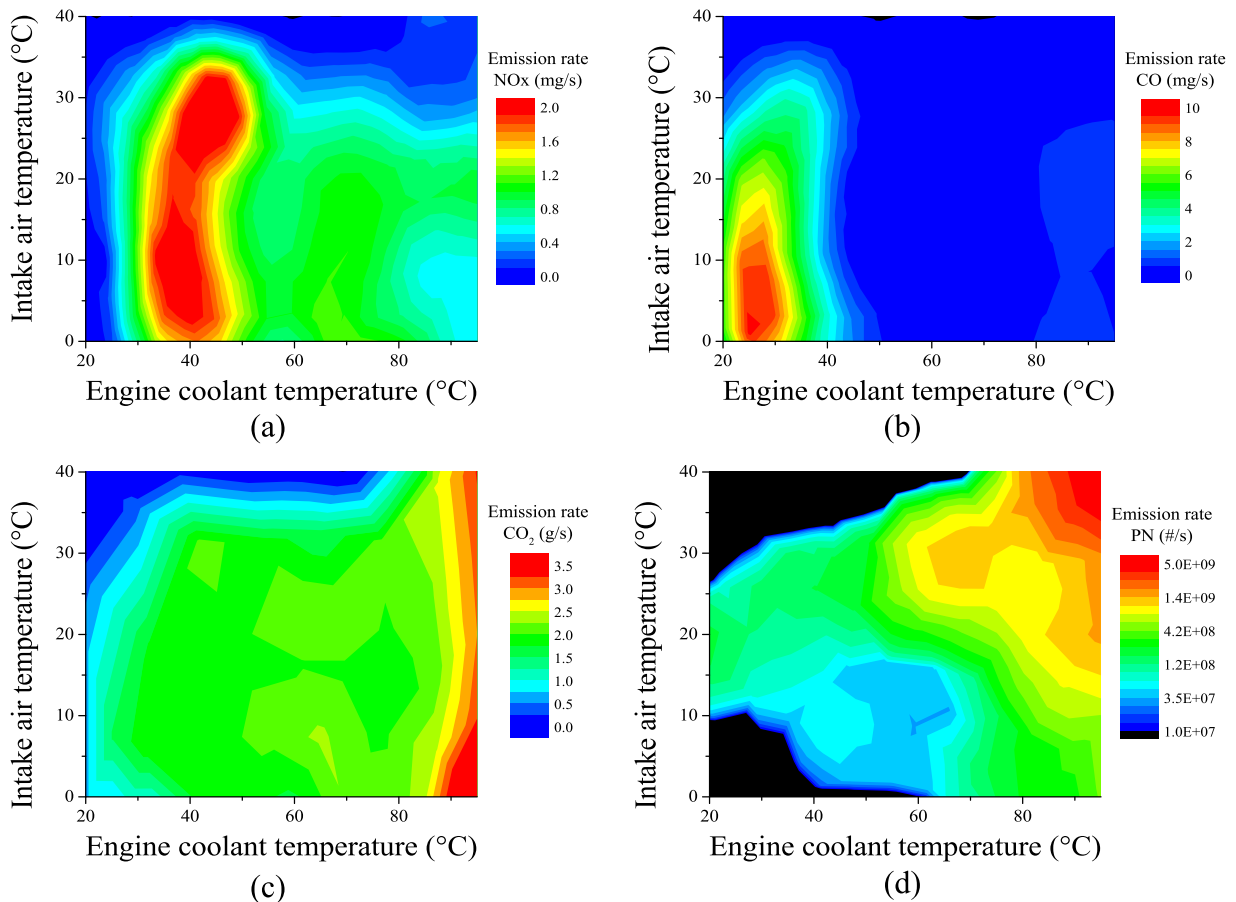


Fig. 5. Exhaust emissions according to the engine coolant temperature and intake air temperature: (a) NOx; (b) CO; (c) CO₂; (d) PN.

understanding the overall trend of NOx emissions with satisfactory accuracy, such as identifying the segments where NOx emissions predominantly occur. These findings demonstrated that the proposed ANN model can serve as a useful means for NOx predictions.

Supplementary Figs. S4 and S5 show that the ANN-predicted NOx emissions are similar to the experimental NOx emissions. The ANN thus showed satisfactory accuracy in NOx prediction. Although the ANN could not perfectly predict the amount of NOx emitted, it accurately predicted the timing of NOx emission increases and decreases. By using seven vehicle operating parameters as model inputs (vehicle speed, engine speed, engine coolant temperature, fuel-air equivalence ratio, exhaust gas temperature, intake air mass flow, intake air temperature) for training, the ANN was able to consider various vehicle and environmental conditions in its emission prediction.

3.2.2. CO emissions

As explained in Section 3.1, the emission map showed that CO emissions are high when the fuel-air equivalence ratio is high and the engine coolant and intake air temperature are low. Those emission characteristics are also visible in Supplementary Figs. S6 and S7. CO emissions were concentrated in a specific area, rather than evenly distributed over the entire period, and they depended considerably on the starting condition. Supplementary Fig. S6(c), (d), and (e) show that large amounts of CO were emitted in the cold start condition. Typically, during the cold start phase, CO emissions increase due to incomplete combustion and the degraded performance of aftertreatment devices (Bielaczyc et al., 2011; Dardiotis et al., 2013). Supplementary Fig. S7(c), (d), and (e) illustrate that a short period after starting accounts for a significant portion of the total CO emissions. Under cold start conditions, CO emissions were higher when the ambient temperature was low. In the three cold-start tests (Test3, 4 and 5), CO emissions within 300 s after the start of the test were higher when the ambient temperature was low (Test 3 CO emissions: about 1.5 g) than when it was high (Test 4 and 5 CO emissions: <0.5 g). In contrast, no sharp increase in CO emissions was observed after starting in the warmed-up condition. As shown in Supplementary Fig. S7(a) and (b), CO emissions increase gradually in the warmed-up condition. The ANN accurately predicted the effects of a cold start on CO emissions, as shown in Supplementary Figs. S6 and S7. Because temperature-related variables such as engine coolant temperature and intake air temperature were used for the machine learning, the ANN was able to estimate temperature-related emission characteristics. In addition to the cold start effect, the ANN well predicted CO emissions in other areas.

3.2.3. CO₂ emissions

Since CO₂ emissions are directly proportional to fuel consumption, they exhibit a strong correlation with kinematic variables such as engine speed and vehicle speed. Our ANN model takes these kinematic variables into account when predicting emissions, resulting in notably high accuracy in CO₂ emission predictions. As depicted in Supplementary Figs. S8 and S9, the comparison between the test data and the ANN-predicted results reveals that the ANN model accurately estimate the CO₂ emissions, encompassing the timing of increases and decreases, as well as the absolute emission levels. This signifies that the ANN model proposed in this study has extensively assimilated the CO₂ emission characteristics. Furthermore, these outcomes indicated the potential of the proposed model in this study can potentially replace conventional hard-sensor-based CO₂ measurement systems.

3.2.4. PN emissions

Supplementary Fig. S10 shows the instantaneous PN emissions during the five on-road test runs. Most instantaneous PN emissions were distributed between 10⁶ and 10⁹ particle/s. In contrast to gaseous emissions, for which we used a linear scale to express the emission rates, we used a logarithmic scale for the PN emission rates. In the cold start conditions shown in Supplementary Fig. S10(c), (d), and (e), high PN emissions (higher than 10⁹ particles/s) immediately after starting appear only in Fig. S10(c). In contrast, the effect of low ambient temperature on PN emissions was more significant in Supplementary Fig. S10(a) and (c). PN emissions above 10⁹ particles/s were measured only under cold ambient temperature conditions. In addition, exceptionally high PN emissions (above 10¹⁰ particles/s) appear in Fig. S10(c).

The correlations between the measured and predicted PN emissions are illustrated in Supplementary Fig. S11. The x-axis and y-axis represent the measured and predicted PN emissions per sec, respectively. The blue dashed line shows the liner regression of PN emissions. The mean of coefficient of determination (R^2) in a linear regression of the five test runs was 0.620, indicating that the PN prediction accuracy of the ANN was satisfactory. In Supplementary Fig. S12, which compares the measured and predicted cumulative PN emissions, the emissions predicted by the ANN are similar to the measured PN emissions.

On the contrary to NOx, CO and CO₂ emissions, the logarithmic transformation applied to PN led to slight distortions in PN emission accuracy. For example, the logarithmic transformation allows a ten-fold difference in PN emissions to be perceived as a unit difference, resulting in a diminished impact of high PN emissions compared to low PN emissions during ANN training. This arises from the inherent difficulty of applying a linear scale to PN emissions, in contrast to NOx, CO, and CO₂ emissions.

Table 2 shows the R^2 and RMSE values between the measured and predicted emissions in all cases. The R^2 for all five test runs for CO₂ prediction is the highest (0.986), and that of PN prediction is the next highest (0.620). The R^2 for all five test runs for CO is 0.236, which is the lowest of the four emission types. The standard deviation of R^2 for the five tests shows the accuracy deviation for each prediction case. The low standard deviation of R^2 for CO₂ emission indicates that the ANN can reliably predict the actual CO₂ emission, regardless of the experimental condition. When the standard deviation of R^2 is large, on the other hand, the ANN performance depends on the test case and is unstable. Although the R^2 for NOx emissions (0.411) is lower than that for PN emissions (0.620), the standard deviation of R^2 for NOx emissions (0.0437) is lower than that for PN (0.1892). Thus, predicting NOx emissions using the ANN is more stable than predicting PN emissions. The RMSE of NOx, CO, CO₂ and PN are 1.66 mg/s, 2.41 mg/s, 249 mg/s, and 7.17·10⁸ #/s, respectively. These values correspond to 2.4%, 1.7%, 1.4%, and 1.4% relative to the maximum measured emission rates for NOx, CO, CO₂, and PN, respectively.

3.3. Emission prediction accuracy according to driving conditions

3.3.1. Emission predictions by vehicle speed

We divided vehicle speed into three parts according to the driving route for the RDE regulation (Park et al., 2021): urban (vehicle speed ≤60 km/h), rural (60 km/h < vehicle speed ≤90 km/h), and motorway (90 km/h < speed).

Table 3 shows the comparison between the measured and ANN-predicted emissions according to the driving conditions. The emission prediction accuracy for NOx emissions in the urban phase (R^2 : 0.43, error: 18.0%) is higher than in the rural (R^2 : 0.37, error: 26.0%) and motorway phases (R^2 : 0.198, error: 56.1%). The time-based emission comparisons in Supplementary Fig. S13 also show that the NOx prediction accuracy is high in the urban and rural phases and low in the motorway phase. This result indicates that the ANN was properly trained for NOx emission characteristics in urban and rural driving conditions. Although the NOx prediction accuracy in the motorway phase is low, the overall NOx prediction accuracy is high because the effects of NOx emissions in the motorway

Table 2

R^2 and RMSE values between measured and ANN-predicted emissions for the five tests.

	R^2 / RMSE (mg/s for NOx, CO, and CO ₂ , #/s for PN)			
	NOx	CO	CO ₂	PN
Test 1	0.356 / 2.73	0.288 / 1.65	0.986 / 324	0.391 / 2.31·10 ⁸
Test 2	0.515 / 1.46	0.212 / 0.39	0.980 / 226	0.784 / 3.63·10 ⁷
Test 3	0.471 / 1.47	0.302 / 4.38	0.984 / 274	0.282 / 1.45·10 ⁹
Test 4	0.394 / 0.71	0.751 / 1.52	0.993 / 194	0.699 / 7.63·10 ⁷
Test 5	0.464 / 0.54	0.513 / 0.61	0.991 / 140	0.621 / 1.64·10 ⁸
All	0.411 / 1.66	0.236 / 2.41	0.986 / 249	0.620 / 7.17·10 ⁸
Standard deviation of R^2	0.0437	0.1463	0.0023	0.1892

Table 3
Emission prediction accuracy of ANN according to vehicle speed and temperature.

Driving conditions	Measured / predicted emissions				R ²			
	NOx (g/km)	CO (g/km)	CO ₂ (g/km)	PN (#/km)	NOx	CO	CO ₂	PN
Urban (speed ≤60 km/h)	0.0857 / 0.0713 (Error: 18.0%)	0.0595 / 0.0398 (Error: 33.1%)	224.0 / 224.7 (Error: 0.31%)	8.14·10 ⁹ / 6.20·10 ⁹ (Error: 23.8%)	0.430	0.134	0.981	0.648
Rural (60 km/h < speed ≤90 km/h)	0.0163 / 0.0205 (Error: 26.0%)	0.0237 / 0.0166 (Error: 30.0%)	147.8 / 147.2 (Error: 0.39%)	6.01·10 ⁹ / 3.03·10 ⁹ (Error: 49.7%)	0.370	0.335	0.985	0.039
Motorway (90 km/h < speed)	0.00288 / 0.00449 (Error: 56.1%)	0.0425 / 0.0276 (Error: 35.1%)	164.4 / 163.6 (Error: 0.48%)	6.81·10 ⁹ / 4.23·10 ⁹ (Error: 37.9%)	0.198	0.369	0.984	0.094
Cold operation (Engine coolant temp. ≤ 80 °C)	0.108 / 0.102 (Error: 5.2%)	0.193 / 0.157 (Error: 18.8%)	282.1 / 279.9 (Error: 0.81%)	2.15·10 ¹⁰ / 2.01·10 ¹⁰ (Error: 6.63%)	0.508	0.454	0.976	0.671
Warmed-up operation (80 °C < Engine coolant temp.)	0.0428 / 0.0372 (Error: 13.2%)	0.0393 / 0.0252 (Error: 35.9%)	183.6 / 183.6 (Error: 0.02%)	6.63·10 ⁹ / 4.20·10 ⁹ (Error: 36.7%)	0.409	0.208	0.987	0.617
Total	0.0453 / 0.0396 (Error: 12.5%)	0.0452 / 0.0302 (Error: 33.1%)	187.3 / 187.3 (Error: 0.02%)	7.20·10 ⁹ / 4.81·10 ⁹ (Error: 33.2%)	0.411	0.236	0.986	0.620

phase (0.00288 g/km) are insignificant across the entire driving route (0.0453 g/km) as shown in Table 3. Supplementary Fig. S13 demonstrates that the cumulative NOx emissions during the motorway phase are <2% of the total on-road NOx emissions.

The CO emissions show a different tendency from the NOx emissions. The R² for CO prediction in the urban phase (R²: 0.134) is lower than that in the rural (R²: 0.335) and motorway phases (R²: 0.369). The low R² value in the urban phase is due to the unpredictable high CO emission rates shown in Supplementary Fig. S14. These steep and high CO emissions were not predicted by the ANN. However, when we recalculated the R² value for the urban phase without the high CO emission rate (10,000–12,000 s in Supplementary Fig. S14), the R² for the urban phase was similar to the that for the rural and motorway phases. The cumulative CO emissions by vehicle speed are illustrated in Supplementary Fig. S15. The cumulative error in CO emissions was about 30% in the urban, rural, and highway phases, and no significant difference depending on vehicle speed was observed.

The CO₂ prediction accuracy is high in all cases regardless of vehicle speed. Both the R² and prediction error are similar for the urban, rural, and motorway phases: urban (R²: 0.981, error: 0.31%) rural (R²: 0.985, error: 0.39%), and motorway (R²: 0.984, error: 0.48%). The cumulative CO₂ emissions shown in Supplementary Fig. S16 illustrate that the error between the measured and ANN-predicted CO₂ emissions is negligible. These results indicate that the ANN can accurately predict CO₂ emissions across all vehicle speed ranges.

For PN emissions, the R² for the urban phase (0.648) is higher than for the rural (0.039) and motorway phases (0.094), which is similar to the tendency in the NOx prediction results. The emission prediction error in the urban phase (23.8%) is also lower than in the other phases (rural: 49.7%, motorway: 37.9%). In high-speed driving, several points of high PN emission rates could not be predicted by ANN, resulting in low accuracy in the rural and highway phases.

ANNs have generally well predicted exhaust emissions, but they have not been able to accurately predict sudden spikes in emissions as shown in Supplementary Fig. S14. Such sudden high emissions usually occur in extreme conditions such as rapid acceleration and high load operation. Therefore, in order to increase the prediction accuracy for sudden high emission, it is necessary to train the ANN using sufficient emission test data measured in extreme environment.

Based on a comparison of emission prediction results by speed, the accuracy of ANN predictions was higher in the urban phase than in the rural and motorway phases, excluding CO emissions. Generally, the urban area, due to its high population density, is a crucial zone for emission management. The relatively higher prediction accuracy of the ANN in the urban phase suggests that ANN-based emission prediction methods can be an effective tool for air quality management in urban areas.

3.3.2. Emission predictions by engine temperature

The thermal conditions of an engine significantly affect exhaust emissions. If the engine is not warmed up sufficiently, pollutant emissions increase due to incomplete combustion and the low efficiency of aftertreatment devices (Weilenmann et al., 2009). Engine coolant temperatures below 80 °C were classified as cold operation, and those above 80 °C were classified as warmed-up operation. In cold start conditions, as shown in Fig. 1(b), engine coolant temperature begins at the ambient temperature and stabilizes at 80–100 °C. After a cold start, it usually takes 10–15 min for the engine to warm up sufficiently. In this study, Tests 3, 4, and 5 were conducted in cold start conditions, as shown in Fig. 1(b).

Table 3 shows a summary of the emission prediction results. CO emissions were more affected by temperature than the other emissions. In the measured emissions, CO emissions generated during cold operation (0.193 g/km) were five times higher than those generated during warmed-up operation (0.0428 g/km). In contrast, CO₂ emissions in cold operation (282.1 g/km) were only about 1.5 times higher than in warmed-up operation (183.6 g/km), which is the lowest rate of increase among the four emission types.

The comparison shows that the ANN accurately predicted cold start emissions. For NOx, CO, and PN, the emission prediction accuracy for cold operation was slightly higher (higher R² and lower error) than the emission prediction accuracy for warmed-up operation. For CO₂ emissions, the ANN showed significantly high accuracy in both cold and warmed-up operation conditions. Supplementary Fig. S18 shows the cumulative emissions in cold and warmed-up operations. No significant difference in NOx, CO₂, or PN emissions was observed between cold and warmed-up operation. In contrast, CO emissions were greatly affected by temperature. In the warmed-up condition shown in Supplementary Fig. S18(c), CO emissions increased gradually over time. In the cold start conditions

shown in Supplementary Fig. S18(d), a large amount of CO was emitted immediately after engine start in Tests 3, 4, and 5, mainly due to the low temperature of the engine.

4. Conclusion

In this study, emission maps proved to be effective tools for analyzing overall emission characteristics. The Type 1 emission map indicated that CO₂, NO_x, and CO emission rates increased as vehicle speed increased. However, the PN emission rate showed no specific tendency. On the other hand, the Type 2 emission map shows emission rates as a function of engine speed and fuel-air equivalence ratio. As the engine speed and fuel-air ratio increased, the CO₂ emissions increased gradually. NO_x emissions were more sensitive to engine speed than the fuel-air equivalence ratio, while CO and PN emissions were highly sensitive to the fuel-air equivalence ratio. Finally, the Type 3 emission map displayed the effect of thermal conditions on exhaust emissions, with NO_x and CO emissions high at low engine coolant and intake air temperatures.

We designed ANNs to predict instantaneous exhaust emissions using seven vehicle operating variables. By considering various vehicle operating parameters in their emission predictions, the ANNs accurately predicted sharp increases in exhaust emissions under various conditions, such as cold start, high-load operation, and low ambient temperature. The prediction accuracy of each ANN was validated and quantitatively analyzed using R² and standard deviation values. The ANNs accurately predicted CO₂ emissions under all driving conditions, but the prediction accuracy for NO_x and PN depended on the vehicle speed, with more accurate predicted emissions in urban driving conditions than in motorway driving conditions. CO emissions were significantly affected by engine temperature, with a large amount of CO emitted immediately after a cold engine start. The proposed ANN models accurately predicted exhaust emissions under both cold and warmed-up conditions.

In this study, we developed a practical model using readily available variables of vehicles. Consequently, we anticipate that this approach will serve as a foundation for establishing a quantitative emission measurement system, extending the research to cover a more extensive range of vehicles. The development of an accurate PN prediction model is the main novelty of this study, which is difficult to estimate using existing road vehicle emission models. However, future studies should focus on the particle sizes of exhaust emissions as we did not consider the particle size distribution in this study. Furthermore, from the perspective of optimizing the performance of ANN, there is a need for research on different types of ANN models, such as RNN, LSTM, or hybrid architectures, while also focusing on optimizing the ANN structure for different types of emissions.

CRedit authorship contribution statement

Jigu Seo: Writing – original draft, Investigation. **Yunsung Lim:** Conceptualization. **Jungwon Han:** Investigation. **Sungwook Park:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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